

# 254 ENGINEERING SERVICES

## Roof Certification Protocol

| Document        | Value                                                                    |
|-----------------|--------------------------------------------------------------------------|
| Title           | Roof Certification Protocol for Existing Roofs                           |
| Document number | 254-RC-001                                                               |
| Version         | 1.1                                                                      |
| Issue date      | September 18, 2026                                                       |
| Prepared by     | Aman Dhakal, P.E., Engineer of Record, 254 Engineering Services          |
| Approval        | Aman Dhakal, P.E., Engineer of Record                                    |
| Applies to      | All roof certification engagements performed by 254 Engineering Services |
| Supersedes      | Version 1.0, issued September 14, 2026.                                  |

### 1. Purpose

This protocol establishes how 254 Engineering Services performs roof certifications of existing roofs so that the engineer can issue a sealed certification letter supported by an evidence package collected by a field technician under the engineer's supervision.

The engineer signs and seals the certification letter personally and the recipient relies on that opinion. Every requirement in this protocol exists so that the engineer can reach a determination from the evidence package alone.

### 2. Scope

This protocol applies to condition assessment and certification of existing roofs on residential and light commercial structures where a client requires a professional opinion on the condition of the roof for insurance binding or renewal, a property sale, lending, or a comparable purpose recorded at intake.

This protocol does not cover windstorm inspections of ongoing construction, which are covered by the firm's windstorm inspection protocol; structural design; plan review for permitting; or repair specification. Each of those is a separate service line under its own protocol and engagement terms. The firm does not perform forensic or cause-of-loss investigation.

### 3. Governing requirements

| Authority                                                           | Application                                                                           |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Texas Engineering Practice Act, Texas Occupations Code Chapter 1001 | Practice of engineering, responsible charge, and sealing of the certification letter. |
| TBPELS rules, 22 TAC Chapter 137                                    | Responsible charge over work performed by others, sealing, and professional conduct.  |

| Authority                                                           | Application                                                                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 2024 International Residential Code, Chapter 9                      | Reference standard for roof covering, underlayment, flashing, fastening, and attic ventilation on residential structures. |
| 2024 International Building Code, Chapter 15                        | Reference standard for roof assemblies on light commercial structures.                                                    |
| Manufacturer installation instructions and published warranty terms | Basis for exposure, fastening, and ventilation requirements for the covering as installed.                                |
| TDI product evaluations                                             | Where a covering or component wind rating is identified and relied on.                                                    |
| Client criteria recorded at intake                                  | Insurer, lender, or purchaser requirements that the letter must address.                                                  |

No state program certifies the condition of an existing roof in Texas. The certification letter is the engineer's professional opinion of observed condition as of the inspection date. It is not a warranty, a guarantee of future performance, or a representation that the roof will not leak.

## 4. Roles and responsibilities

| Role                     | Responsibility                                                                                                                                                                                                         | Holder                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Engineer of Record (EOR) | Approves this protocol and its appendices. Sets acceptance criteria and thresholds. Reviews every evidence package, records the determination, and signs and seals the certification letter. Does not attend the site. | Aman Dhakal, P.E.                   |
| Field Technician         | Collects evidence on site in accordance with Appendix B under the engineer's supervision. Records observations and limitations. Makes no determination of condition and offers no opinion to the customer.             | Personnel qualified under Section 5 |
| Coordinator              | Intake, job file, scheduling, dispatch, delivery of the letter, and written communication with the customer.                                                                                                           | Robert Reyna or delegate            |

Only the Engineer of Record determines the outcome of a job and the wording of the letter. The field technician and the coordinator relay the engineer's written determination without interpretation, addition, or estimate of when a letter will issue.

## 5. Personnel qualifications

### Field Technician

- At least two years of documented residential or light commercial experience in roofing, construction, or property inspection, or equivalent experience the engineer accepts in writing.
- Completion of the engineer's training on this protocol and Appendix B, with a supervised inspection before independent work.
- Current ladder and roof-access safety training, and fall protection where site conditions require it.
- Annual refresher training when the adopted code, this protocol, or Appendix B changes.

## **Engineer of Record**

- Active Texas P.E. license. A copy, with expiration date, is held in the firm's compliance file.

## **6. Intake**

- The coordinator opens a job file for each property. The file records the intake answers and customer uploads required by Appendix A.
- The engineer records the purpose of the letter and the recipient exactly as given at intake. The letter is addressed only to that recipient and issued only for that purpose.
- A yes to any flag question in Appendix A routes the job to the engineer before dispatch. The engineer records accept, accept with conditions, or decline.
- A job is not dispatched until intake is complete and every upload required by Appendix A has been received.
- The engineer records any condition of acceptance in the job file, including any item the technician must capture beyond Appendix B.

## **7. Scheduling and dispatch**

- The coordinator schedules an accepted job within two business days. The engineer may set a different interval for a job in the job file.
- The technician confirms access, occupancy, and attic availability before travel.
- Where site conditions prevent complete capture, including weather, access, safety, or roof steepness, the technician records each item that could not be observed and the reason. No item is estimated, assumed, or left blank.
- Where the roof cannot be safely walked, the technician does not walk it. The engineer decides from the evidence collected whether a determination is possible or the job requires lift or equipment pricing.

## **8. Field inspection**

The technician completes every item of Appendix B for the property, in the order listed, and follows the photo procedure printed at the head of that checklist.

- Before starting, the technician makes a rough sketch of the roof, marks North, and photographs the sketch.
- The technician ticks the item being photographed, photographs the checklist, then takes the photographs for that item, and moves to the next item the same way.
- An item that does not apply to the property is marked with the reason it does not apply.
- The counts called for on the checklist are recorded as counts, not as descriptions.
- The seal-bond check is performed and the ambient temperature at the time of the check is recorded, because the result is not meaningful without it.

## **9. Evidence standard and engineer review**

### **Evidence package**

Each job produces one evidence package in the firm's field application, consisting of the completed Appendix B checklist, the photographs, and the technician's notes.

- Every checklist item has at least one wide photograph placing the item on the structure and one close photograph showing its condition.
- Where a dimension, exposure, or thickness governs, a ruler or tape is in frame.
- The worst condition on each roof plane is photographed. A representative sample selected for appearance is not acceptable.
- Every patch, prior repair, penetration, and flashing location is photographed individually.
- Photographs carry location and three time values from the field application: the time reported by the device, the server time at sync, and the difference between them. A device clock that disagrees with the server is recorded as disagreeing rather than presented as certain. Photographs from other devices are not accepted.
- An observation that cannot be made is recorded as not observed with the reason.
- A package that does not cover every checklist item, with the photographs each item calls for, is rejected without review.

### **Engineer review**

- The engineer reviews each evidence package and records one of the five determinations in Appendix C: pass, revise, repairs required, site revisit, or decline.
- The engineer's review record identifies the checklist items and photographs relied on for the determination.
- A package that does not allow the engineer to verify a required item is returned for revision or a site revisit. The customer is not charged for a re-inspection caused by an incomplete package.

## **10. Repairs and deficiencies**

- On a repairs required determination, the engineer issues a written repair list identifying each item, what governs it, and what is required to close it.
- Repairs are performed by others. 254 Engineering Services does not perform, supervise, or recommend a contractor for the repair work.
- Corrections are verified on a site revisit to the same evidence standard, and the engineer closes each item individually.
- No certification letter is issued with an open repair item. There is no conditional certification.

## **11. Certification letter**

- On a pass determination, the engineer issues a sealed letter addressed to the recipient recorded at intake, for the purpose recorded at intake.
- The letter identifies the property, the date of inspection, the covering type, the method by which the evidence was collected and reviewed, and the observed condition.
- The letter states observed condition only. It does not estimate remaining service life, forecast future performance, or represent that the roof will not leak.

- The letter states that the inspection was visual and non-destructive, that it reflects condition on the date of inspection only, and that no representation is made about concealed conditions.
- The letter is not reassigned to another recipient, another purpose, or another date. A new inspection is required.
- The engineer does not certify a roof that was not inspected under this protocol by this firm.

## 12. Exclusions

The following are declined at this service tier or routed to another service line under its own engagement terms. The coordinator records the routing in the job file and informs the customer in writing.

- An open insurance claim on the roof, where a certification would function as claim leverage. Declined, with a referral to an independent firm.
- Active or threatened litigation involving the roof.
- A covering type outside the engineer's competency.
- Unsafe access that prevents minimum capture where the customer refuses lift or equipment pricing.
- A customer request for a predetermined conclusion. This is an automatic decline and the request is logged verbatim.

## 13. Records and retention

- Each job file contains: the intake record and customer uploads, the engineer's acceptance record, the dispatch record, the completed Appendix B checklist, all photographs, the technician's notes, the engineer's review record and determination, any repair list and its closure records, and the issued letter.
- Job files are retained for ten years from the date of the letter. For a job that produces no letter, the ten years run from the date of the last engineer determination recorded in the file.
- Ten years is the firm's floor. The engineer reviews that floor against TBPELS rules and the firm's professional liability policy and raises it where either requires longer.
- The compliance file holds the engineer's license records, technician qualification and training records, and every issued version of this protocol.

## 14. Protocol maintenance

- The engineer reviews this protocol annually, and within thirty days of any change to the adopted code, to a governing manufacturer standard, or to the firm's field application.
- Revisions are issued as a new version number with the effective date and the engineer's approval recorded in the document control table.
- This document follows the firm's standard protocol format and carries the portal's four parts as Appendices A, B, and C.

## Appendix A. Intake questions and customer uploads

Asked and collected by the coordinator before the job is created. A job is not dispatched until every required item is present.

## **PART 1 — INTAKE QUESTIONS**

### **Purpose & recipient**

1. What is the letter for? (insurance binding/renewal / home sale / lender / permit / other — record exactly)

Answer: \_\_\_\_\_

2. Who will the letter be addressed to? (insurer name, buyer, lender, city)

Answer: \_\_\_\_\_

3. What deadline is the customer working against, if any?

Answer: \_\_\_\_\_

### **Property basics**

4. Property address and county

Answer: \_\_\_\_\_

5. Property type (single family / duplex / townhome / small commercial) and number of stories

Answer: \_\_\_\_\_

6. Year built; roof age if known; year of last full replacement

Answer: \_\_\_\_\_

7. Covering type (shingle / metal / tile / flat-membrane / built-up / wood / other)

Answer: \_\_\_\_\_

### **Flags — a YES on any of these routes to engineer review BEFORE dispatch**

8. Is there an OPEN insurance claim on this roof? (open claim = engineer pre-review; likely decline)

Answer: \_\_\_\_\_

9. Is there active or threatened litigation involving the roof? (route to engineer; likely decline)

Answer: \_\_\_\_\_

10. Any active leaks right now? (changes scope from certification to assessment conversation)

Answer: \_\_\_\_\_

11. Has another engineer or inspector already issued an adverse report? (must be disclosed and uploaded)

Answer: \_\_\_\_\_

12. Storm damage within the last 12 months? Date and type (hail/wind)

Answer: \_\_\_\_\_

### **Access & safety (decides dispatch feasibility)**

13. Is attic access available and accessible?

Answer: \_\_\_\_\_

14. Number of Stories and Roof Steepness?

Answer: \_\_\_\_\_

15. Occupied or vacant? Accessible if occupied?

Answer: \_\_\_\_\_

### **Prior paperwork prompt**

16. Do you have: prior roof certs, roofing invoices/warranty, permits, insurance letters, photos of any problems?

Answer: \_\_\_\_\_

## **PART 2 — CUSTOMER UPLOADS (collected at intake, before dispatch)**

### **Required**

- Photo of the front of the property (confirms address/structure)

### **Required if it exists**

- Any adverse report, prior engineer letter, or insurer letter driving this request

### **Collect when applicable**

- Roofing contract/invoice/warranty for the most recent roof work (if work claimed)
- Permits for roof work; HOA approval if applicable
- Customer's own photos of problem areas (if a leak or damage prompted the call)
- Prior roof inspection report (if one exists)

## **Appendix B. Technician on-site checklist**

Completed on site by the technician. The checklist cannot be submitted incomplete.

### **PHOTOS**

- Before starting, make a rough sketch of the roof and mark North. Take a photo of it.
- Tick the item you are about to photograph, take a photo of the checklist, then take the photos for that item.
- Move on to the next item the same way.

### **JOB INFORMATION**

Job / Dispatch No.: \_\_\_\_\_

Property address (full, incl. county): \_\_\_\_\_

Technician name: \_\_\_\_\_

Date / arrival time / departure time: \_\_\_\_\_

Weather during visit: \_\_\_\_\_

Contact on site (name, role, phone): \_\_\_\_\_

☐ Photo of address marker (house number, street sign, or mailbox) [PHOTO]

### **CONTEXT (every job)**

☐ All four elevations of the structure [PHOTO]

☐ Each roof plane overview from eave/ladder [PHOTO]

Covering type / shape (hip-gable-flat-combo) / stories: \_\_\_\_\_

☐ Tree contact or overhang over roof — each occurrence [PHOTO]

Technician notes / readings: \_\_\_\_\_

\_\_\_\_\_

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## COVERING CONDITION

- ☐ Close-up with ruler of covering in GOOD area — exposure/profile baseline [PHOTO]
  - ☐ Close-up of WORST area on each plane [PHOTO]
  - ☐ Shingle roofs — granule loss: close-up of worst granule condition + gutter/downspout granule accumulation [PHOTO]
  - ☐ Shingle roofs — cracking, curling, cupping, blistering: worst example each, wide + close [PHOTO]
  - ☐ Shingle roofs — SEAL-BOND: gentle tab lift at 4 locations spread across different planes; photo any unsealed tab
- Ambient temperature during seal-bond check (°F): \_\_\_\_\_
- ☐ Metal roofs — panel corrosion, seam condition, fastener back-out: worst examples with ruler [PHOTO]
  - ☐ Metal roofs — oil-canning or panel distortion wide shots [PHOTO]
  - ☐ Tile roofs — cracked/slipped/missing tiles: count per plane, worst close-ups [PHOTO]
  - ☐ Flat/built-up — ponding evidence (staining rings), blisters, alligatoring, seam condition, drainage path to scuppers/drains [PHOTO]
  - ☐ ALL patches and prior repairs — every one, wide + close [PHOTO]
  - ☐ Biological growth: moss, algae, dark streaking — worst area each plane affected [PHOTO]
  - ☐ Debris accumulation in valleys, behind chimneys, at wall intersections [PHOTO]
  - ☐ Exposed or backed-out fasteners — worst examples [PHOTO]
- Counts — patches \_\_\_\_ damaged shingles/tiles visible \_\_\_\_ unsealed tabs found \_\_\_\_: \_\_\_\_\_
- Technician notes / readings:
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## EDGES, FLASHINGS, PENETRATIONS

- ☐ Drip edge/eave condition each side; fastening with ruler at 2 spots [PHOTO]
- ☐ Fascia and eave boards: rot, deterioration, separation — full run each side, close-up of worst [PHOTO]
- ☐ Ridge/hip caps condition [PHOTO]
- ☐ Every penetration: boots (cracked rubber = close-up), vents, flues [PHOTO]
- ☐ Skylights: glass cracks, condensation between panes (failed seal), curb flashing — exterior + interior shot each [PHOTO]
- ☐ Sealant condition at every flashing and penetration: dried, cracked, or gapped caulking — close-up each occurrence [PHOTO]
- ☐ Chimney: flashing + masonry condition (loose brick, mortar, cap) [PHOTO]
- ☐ Wall-roof intersections and kick-out flashing at gutter terminations [PHOTO]



☐ Gutters: attachment, separation, standing debris level [PHOTO]

Technician notes / readings:

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### ATTIC VERIFICATION

☐ Deck material + thickness at hatch with ruler [PHOTO]

Deck material / thickness: \_\_\_\_\_

☐ Daylight check: lights off, photo toward each plane — any daylight = close-up at source [PHOTO]

☐ Staining/leak evidence: every stain, wide + close, wet-test note (dry stain vs active moisture) [PHOTO]

☐ Deck sag or delamination between framing — sight along planes, photo any deviation [PHOTO]

☐ Fastener shiners: length close-up at 2+ spots [PHOTO]

☐ Ventilation: intake (soffit) and exhaust (ridge/box/turbine) present — one photo each type [PHOTO]

☐ Soffit vents unblocked: check insulation not covering intake at eaves — photo 2+ eave bays [PHOTO]

☐ Insulation condition: damp, matted, or compacted areas anywhere (not just at leaks) [PHOTO]

☐ Pest evidence: nests, droppings, chewed framing/wiring [PHOTO]

Technician notes / readings:

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### INTERIOR FINISH SCAN

☐ Top-floor ceilings each room: stains, peeling paint, bubbling drywall — wide shot per room, close-up of any finding [PHOTO]

☐ Musty odor noted by room

Technician notes / readings:

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### COVERING AGE AND EXPOSURE

☐ Any covering product label/wrapper/stamp found on site or in attic [PHOTO]

Owner-reported roof age / replacement year (verbatim): \_\_\_\_\_

☐ South-facing plane close-up (fastest-aging plane) with ruler [PHOTO]

☐ Any solar panels, satellite mounts, or rooftop equipment: attachment condition [PHOTO]

Technician notes / readings:

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### **CLOSE-OUT**

☐ Full perimeter walk — anything unusual [PHOTO]

☐ Exception menu cleared for every non-applicable item

Total photos: \_\_\_\_\_

Technician signature / date: \_\_\_\_\_

Technician notes / readings:

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## **Appendix C. Engineer decision rules**

Applied by the engineer to each evidence package. One determination is recorded per review.

### **PASS**

- Package complete: every checklist item present or properly excepted
- No active leak evidence (no daylight, no active moisture at stains)
- Covering condition consistent with reported age; worst-area photos show serviceable covering per the letter's stated criteria
- Seal-bond, fastening, flashings, and penetrations show no condition requiring repair before certification

### **REVISE**

- Any required photo missing, blurry, or missing its ruler/context pair
- Worst-area coverage doubtful (all photos look selectively good)
- Exception used where the condition plainly applied
- Counts inconsistent with photos

### **REPAIRS REQUIRED (certification withheld, repair list issued)**

- Active leak: daylight through deck or active moisture at any stain
- Deck delamination or sag confirmed by measurement
- Failed flashing, boots, or sealant at any penetration
- 2 or more unsealed tabs of the 4 locations tested
- 10 or more damaged shingles or tiles on any plane; for metal, standing-seam, or membrane coverings, any breach of the water barrier
- 8 or more hail hits in a 10 foot by 10 foot (100 square foot) test square, where a test square was taken on a site revisit
- Certification proceeds only after repairs are verified on revisit

### **SITE REVISIT (technician returns with an engineer-specified capture list)**

- Attic coverage incomplete: hatch photo only, no plane-by-plane daylight and stain check
- Stain present without a moisture reading — return with meter and wet-test each stain
- Sag or delamination suspected in sightline photos — return for straightedge/string-line measurement and framing spacing
- Storm damage reported within 12 months — return for test-square count on each plane (hail hits / creased tabs)
- Conflicting evidence (owner story vs photos vs uploads) — return for the specific items that settle it: product label, permit sticker, invoice match to covering
- Prior adverse report exists — return and photograph every condition the report cites at the same locations before contradicting it
- Repairs required — return after repairs to verify each item on the repair list

## DECLINE

- Open insurance claim where the certification would function as claim leverage — decline and refer to an independent firm
- Active or threatened litigation — decline at this service tier
- Covering type outside engineer competency.
- Unsafe access that prevents minimum capture and customer refuses lift/equipment pricing
- Customer requests a predetermined conclusion — automatic decline, log verbatim request

## Appendix D. References

| Item                                                                                 | Location                                                                         |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 2024 International Residential Code, Chapter 9, Roof Assemblies                      | As adopted by the authority having jurisdiction                                  |
| 2024 International Building Code, Chapter 15, Roof Assemblies and Rooftop Structures | As adopted by the authority having jurisdiction                                  |
| Texas Engineering Practice Act, Texas Occupations Code Chapter 1001                  | <a href="http://statutes.capitol.texas.gov">statutes.capitol.texas.gov</a>       |
| TBPELS rules, 22 TAC Chapter 137                                                     | <a href="http://pels.texas.gov">pels.texas.gov</a>                               |
| TDI Product Evaluations                                                              | <a href="http://tdi.texas.gov/wind/index.html">tdi.texas.gov/wind/index.html</a> |
| Manufacturer installation instructions and warranty terms                            | Manufacturer publication for the identified product                              |

## Approval

Reviewed and approved by the Engineer of Record. This protocol is in force from the date below.

Aman Dhakal, P.E.

Engineer of Record, 254 Engineering Services

Signature:  Date: 09/20/2026